

ACM/ICPC Regional Hong Kong 2017

solution

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Problem A

Enumerate x , the y minimizing $F(x, y, z, n)$ is in order.

Problem B

- Scan a line from $x = 0$ to $x = n$
- Use a segment tree to maintain the color of grids on the scan line

Problem C

- Enumerate rotation and flipping
- Use an array to record the bijection

Problem D

Shortest path

Problem E

- Binary search the distance
- Pick the station with a greedy strategy

Problem F

Calculate according to the description

Problem G

Dynamic programming

Problem H

- The integers in a set is continuous
- $S_i = \{x \in \mathbb{I} | 2^i - 1 \leq x \leq 2^{i-1}n - 2^{i-1} + 1\}$
- binary search which set contains the k -th number

Problem I

Calculate the recursive equation

- $F_0 = 0$
- $F_1 = 1$
- $F_{2n} = 3f_n + n(n-1)/2$
- $F_{2n+1} = 2f_n + f_{n+1} + n(n+1)/2$

Problem J

- The beautiful triangles consist of two black points and one white point, or one black point and two white points.
- Each beautiful triangle is only decided by a random edge
- Choose the direction between each pair point with same color with a greedy strategy

Problem J

- Denote $P[i]$ as the number of white points which is in distance D of black point i , and $Q[i]$ as the number of black points which is in distance D of white point i .
- The maximum number of beautiful triangles is
$$\sum_i ((i-1) * (P[i] + Q[i])) - (P[i] * (P[i]-1) + Q[i] * (Q[i]-1)) / 2$$
- The minimum number is similar.
- $P[i]$ and $Q[i]$ can be calculated with a segment tree after rotating all point by 45 degree

Problem K

- Calculate the shortest path between each pair of grids
- Use bfs with status memorized